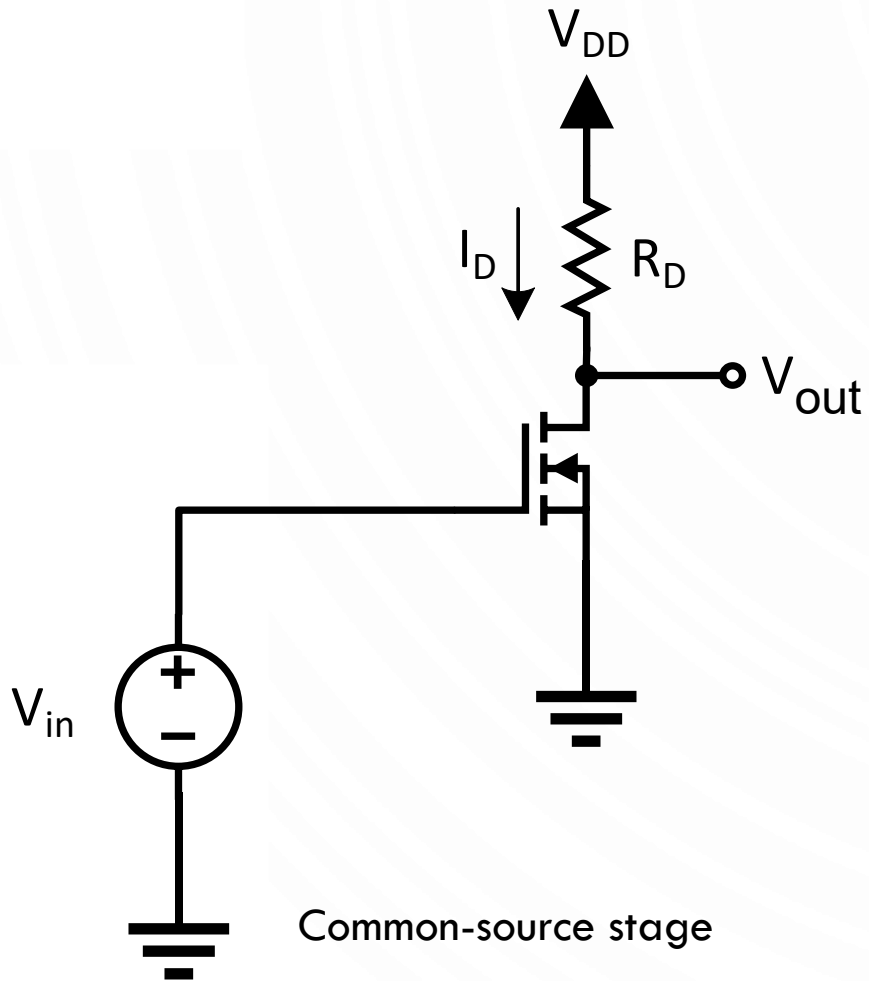


Current Mirrors

Why do we need current mirrors and where do we use them?

CURRENT MIRRORS: GENERAL DEFINITIONS AND PROPERTIES

Scenario 1:



$$I_D = I_{Dsat} = \frac{W \mu_n C_{ox}}{2L} [(V_{GS} - V_T)^2] \quad V_{Dsat} \leq V_{DS}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$A_v = -g_m R_D$$

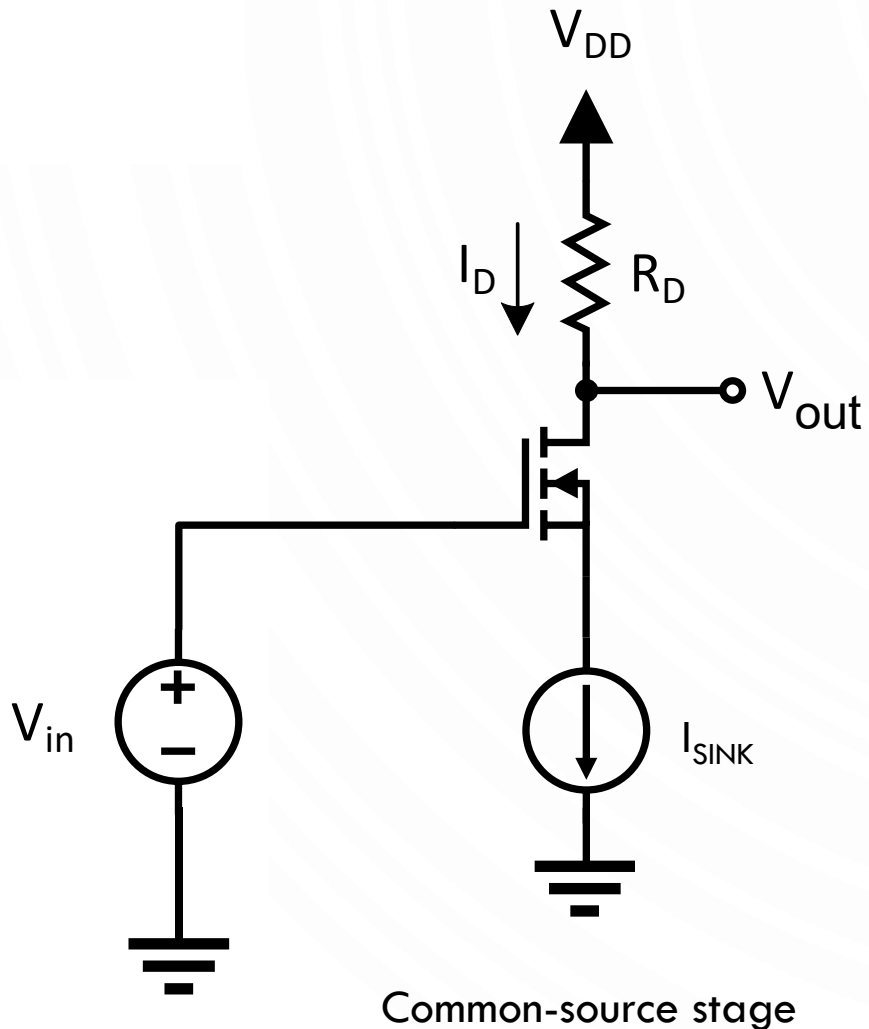
The gain of the common-source amplifier depends on the transconductance g_m and the load resistor R_D .

g_m depends on diverse factors like drain current I_D , temperature etc. If we need a constant gain, we need to keep the current constant.

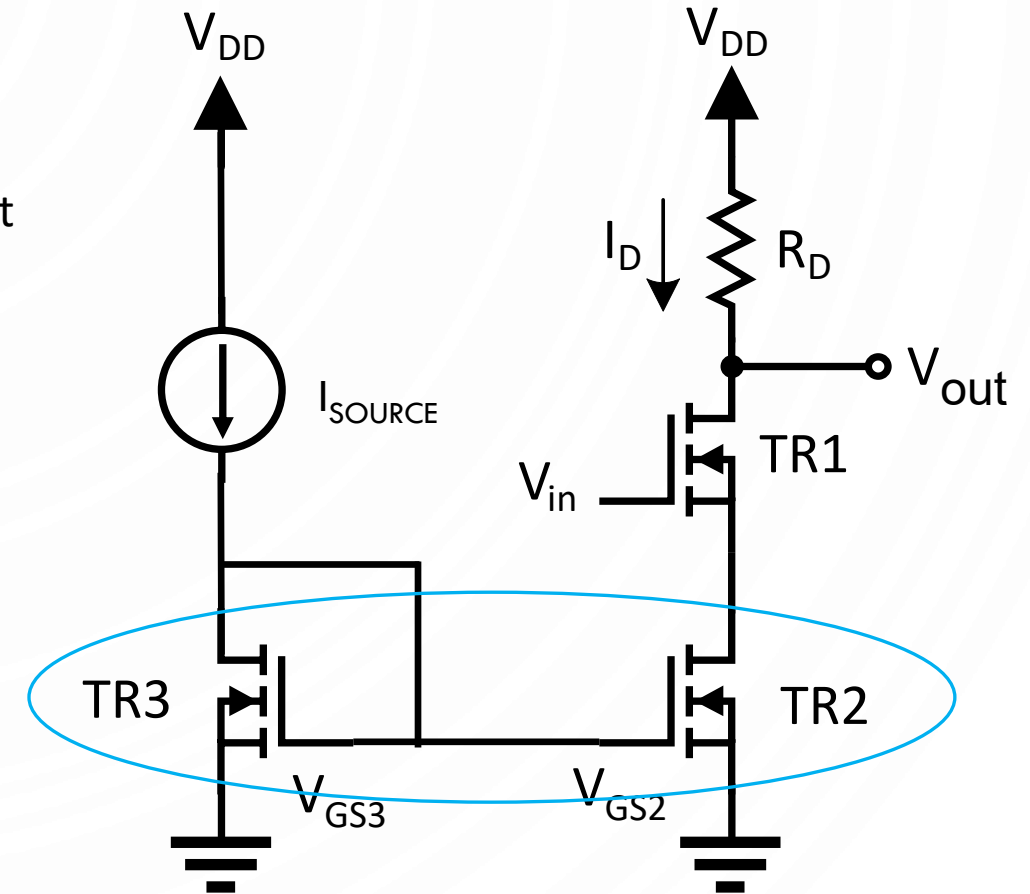
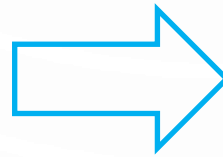
CURRENT MIRRORS: GENERAL DEFINITIONS AND PROPERTIES

Scenario 1:

If we introduce a current source with a constant current of I_{SOURCE} , the variation of g_m due to a change of transistor current is eliminated.



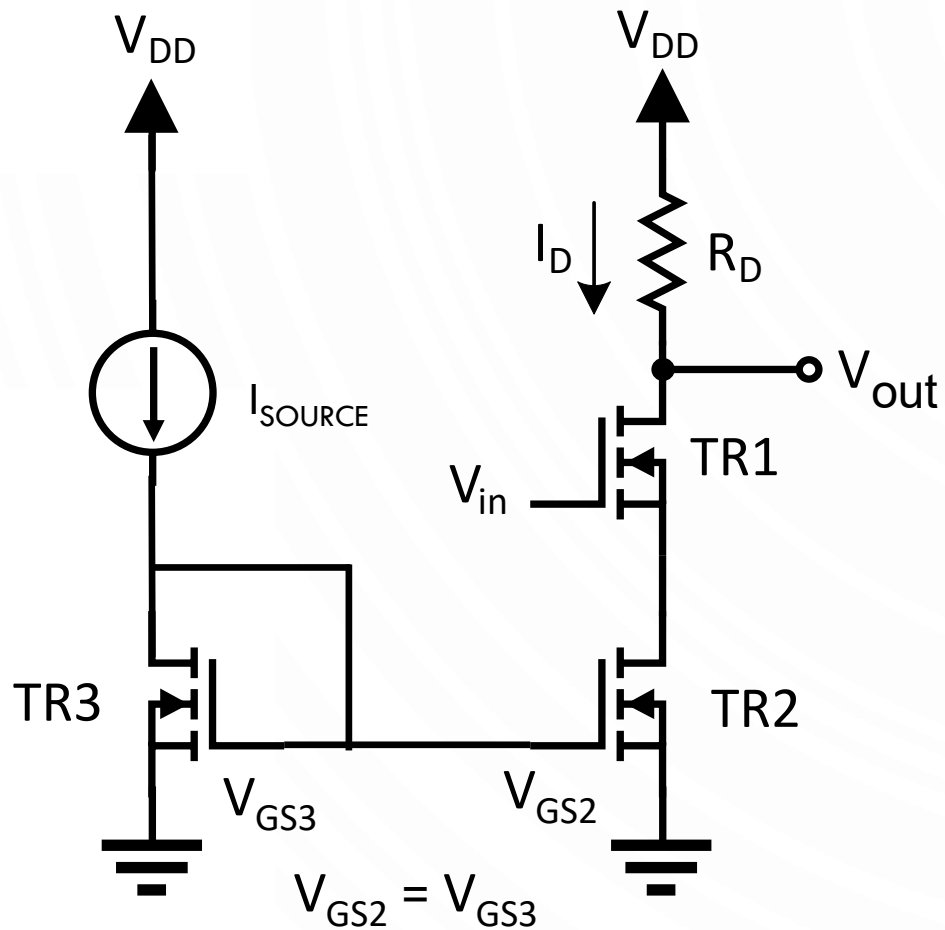
The current source I_{SOURCE} : How do we implement this current source? I_{SOURCE} is essentially a MOSFET.



V_{GS2} and V_{GS3} must be equal according to Kirchhoff's Law!

CURRENT MIRRORS: GENERAL DEFINITIONS AND PROPERTIES

Scenario 1:



Example:

$I_{\text{SOURCE}} = 1 \text{ mA}$, if we want to have a drain current I_D of 4 mA how many parallel TR2 transistors do we need?

Solution: if 4 TR2 transistors are connected in parallel, then the total width W of TR2 is increased by a factor of 4. According to

$$I_D = I_{Dsat} = \frac{W \mu_n C_{ox}}{2L} [(V_{GS} - V_T)^2] \quad V_{Dsat} \leq V_{DS}$$

This equation is valid, if we operate TR2 and TR3 in saturation region.

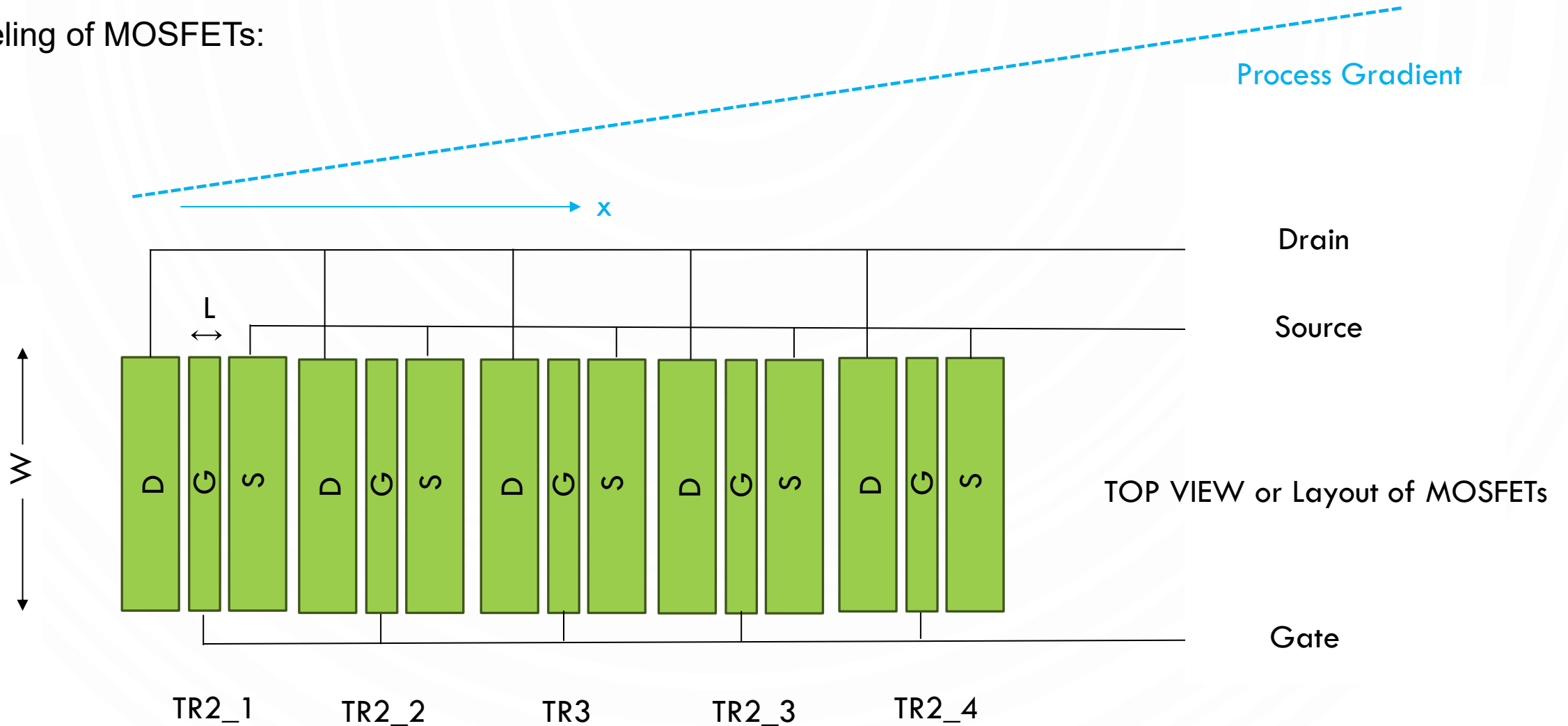
The current flowing through TR2 is x 4 higher (namely equal to 4 mA), than the current flowing through TR3 if they have the same gate lengths. $k_M = 4$.

How is I_{SOURCE} generated? This will be discussed in the following slides/weeks...

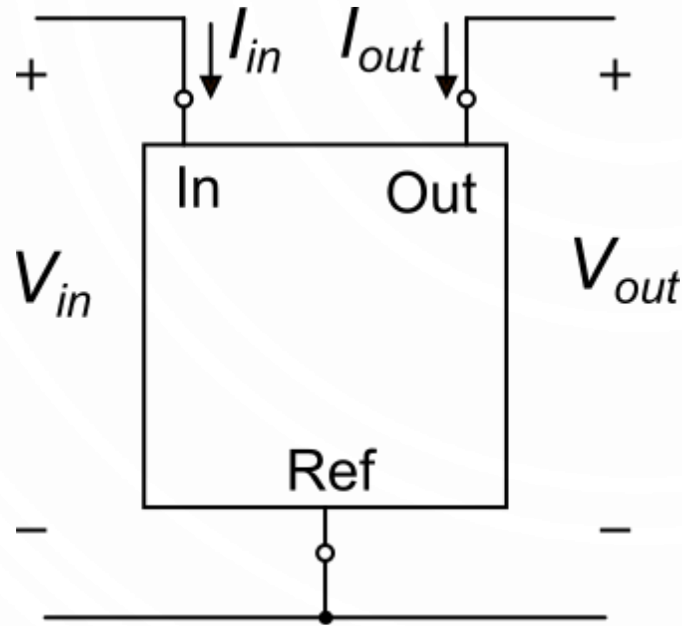
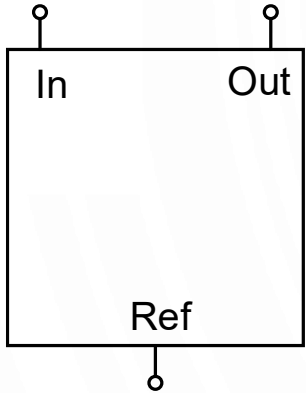
CURRENT MIRRORS: GENERAL DEFINITIONS AND PROPERTIES

Scenario 1: TR2 consists of four parallel connected transistors, TR3 consists of only a single transistor

Paralleling of MOSFETs:



CURRENT MIRRORS: GENERAL DEFINITIONS AND PROPERTIES

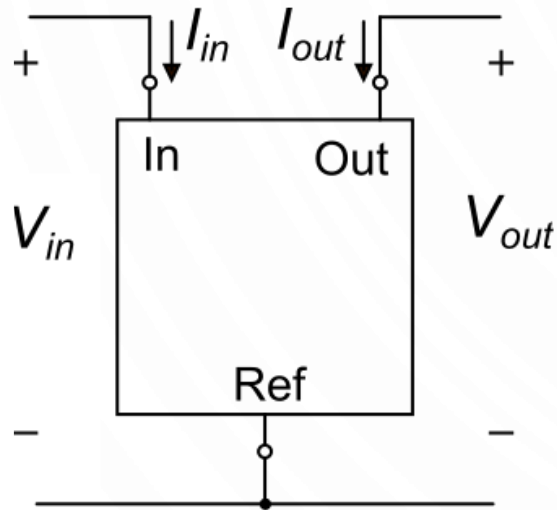


Desired characteristics

$$V_{in} = 0$$
$$I_{out} = k_M I_{in}$$

These characteristics are generally non feasible with simple elementary circuits.

CURRENT MIRRORS: N-TYPE AND P-TYPE MIRRORS



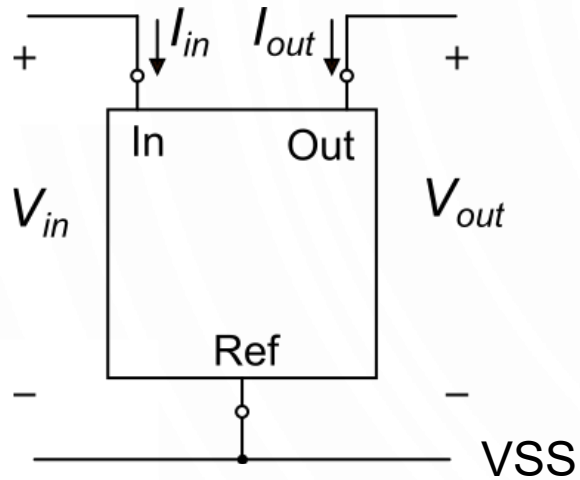
$$I_{out} = k_M I_{in}$$

$V_{out}, V_{in} > 0$
 $I_{out}, I_{in} > 0$ **n-type:**
formed mainly by n-type
transistors

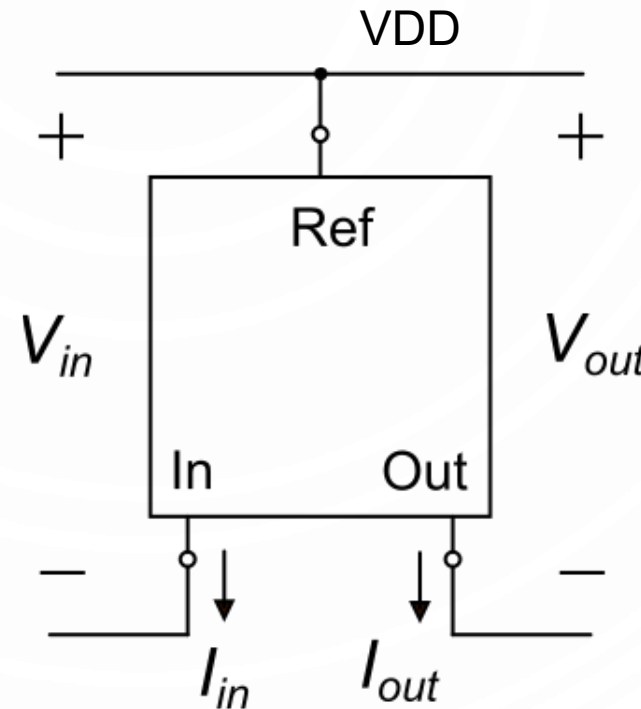
$$k_M > 0$$

$V_{out}, V_{in} < 0$
 $I_{out}, I_{in} < 0$ **p-type:**
formed mainly by p-type
transistors

CONVENTIONS ON VOLTAGE AND CURRENT SIGNS USED TO OBTAIN POSITIVE VALUES ALSO FOR P-TYPE MIRRORS



Convention for an n-type current mirror

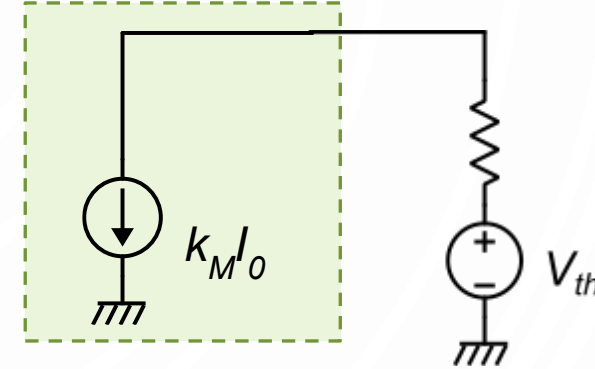
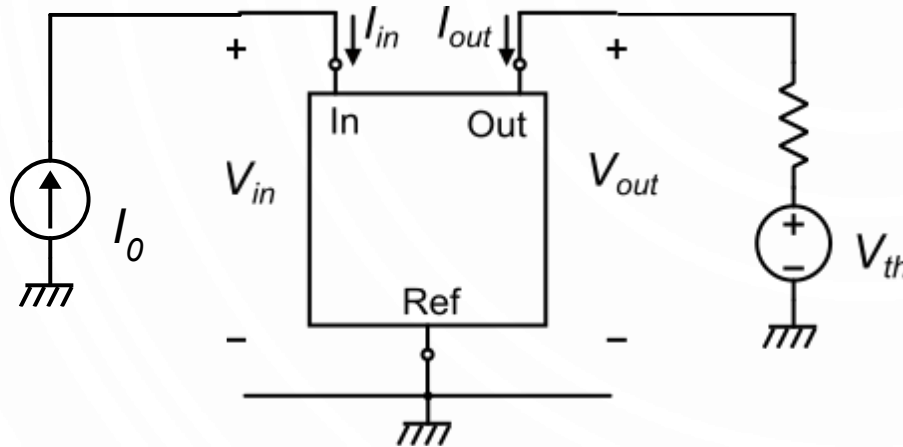


Convention for a p-type current mirror

With this convention all quantities are positive also for the p-type mirror

CURRENT REVERSING FUNCTION OF A CURRENT MIRROR

Equivalent circuit seen from the mirror output terminal



This is a current **source** (the current exits from the generator)

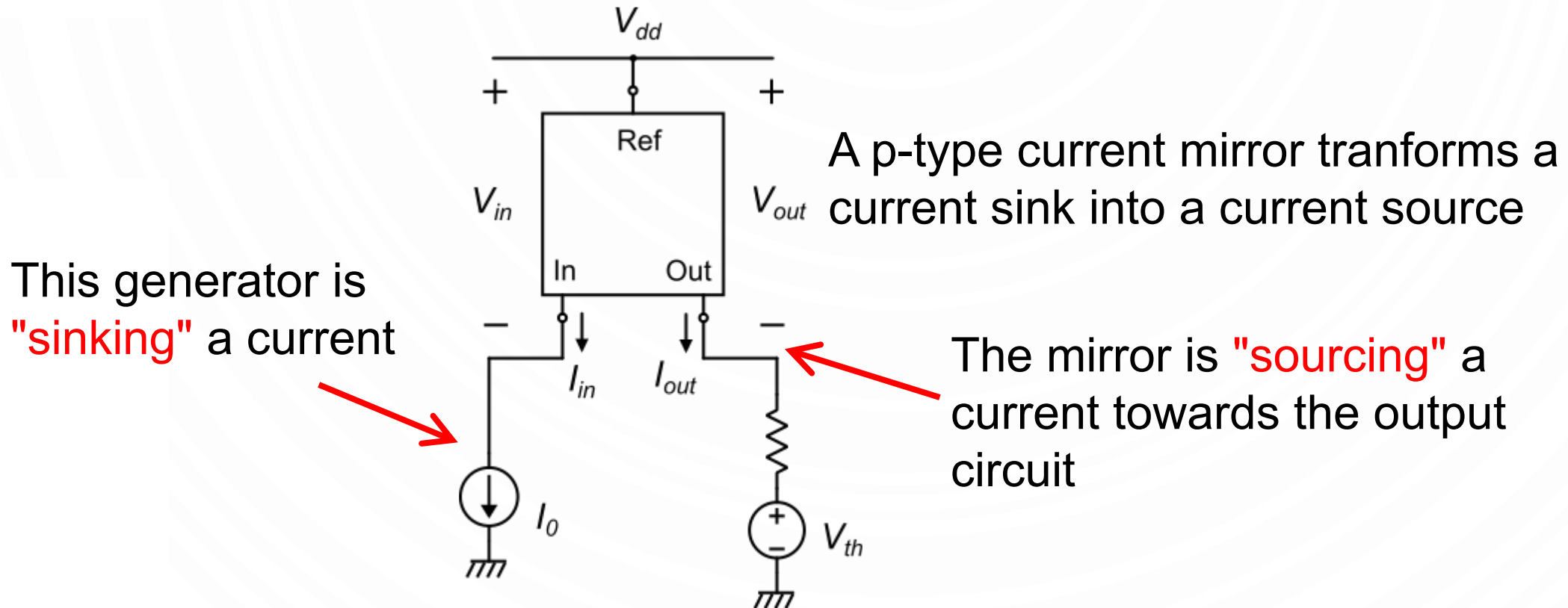
n-type mirror is required (the input current enters the mirror)

The circuit connected to the output port must be able to provide a positive V_{out}

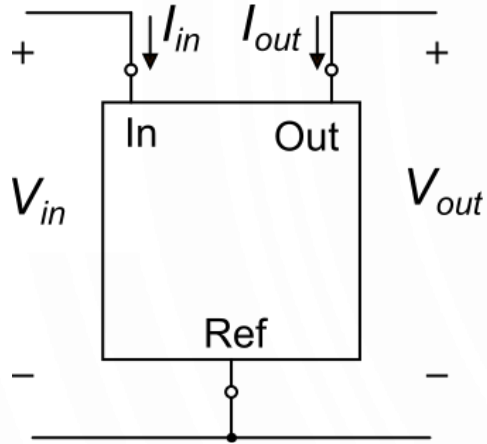
The mirror is seen by the output circuit as an equivalent current source. Note that now the current enters the source (**=sink**)

CURRENT REVERSING FUNCTION OF A CURRENT MIRROR

An n-type current mirror transforms a current source (the current exits from the source) into a current sink (the current enter the source)



PARAMETERS OF MERIT FOR A CURRENT MIRROR

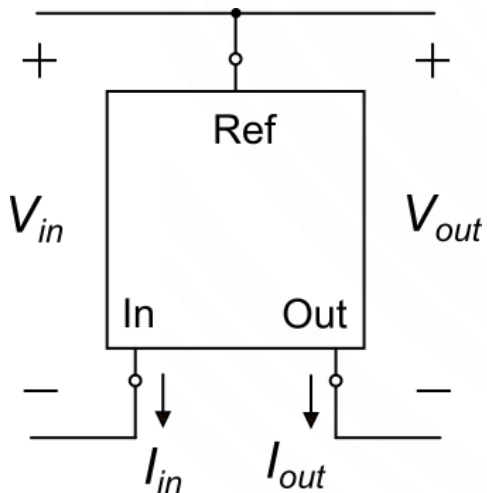


Ideal function: $I_{out} = k_M I_{in}$

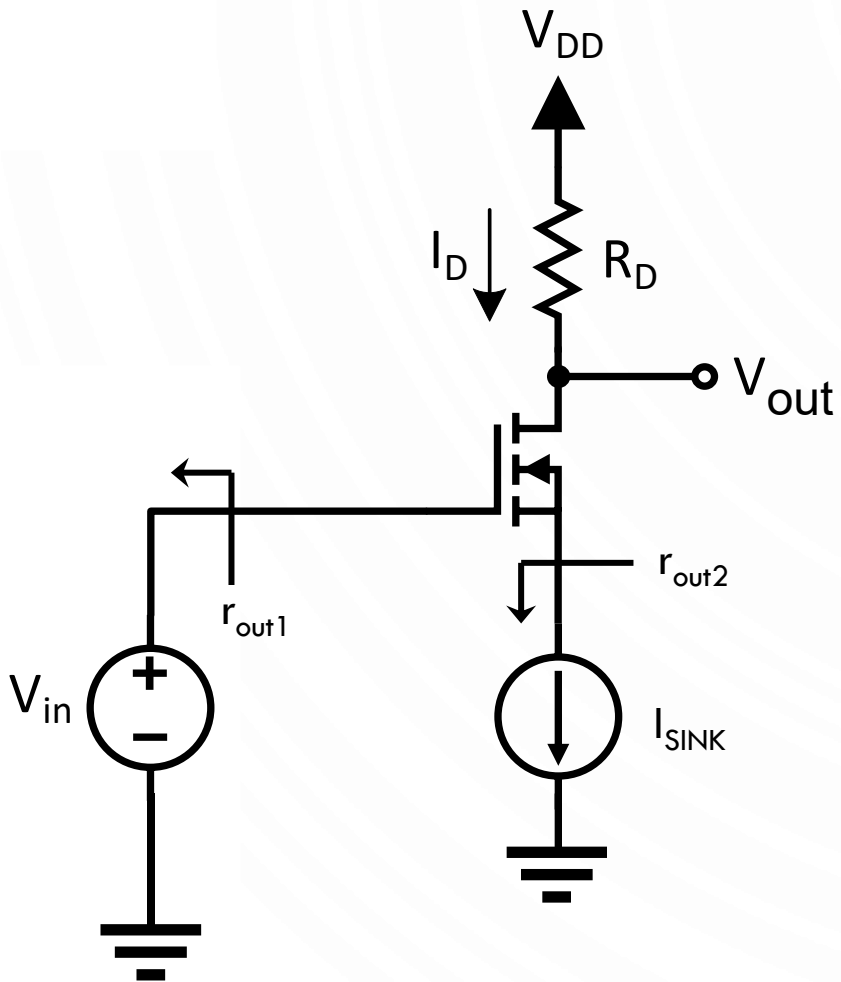
General conditions

$$V_{out}, V_{in} > 0$$
$$I_{out}, I_{in} > 0$$

1. V_{in} should be almost constant and be as small as possible
2. The ideal function should be maintained down to very small V_{out} values. The smallest V_{out} value is indicated with V_{MIN}
3. For $V_{out} > V_{MIN}$, the dependence of I_{out} on V_{out} must be the smallest as possible.



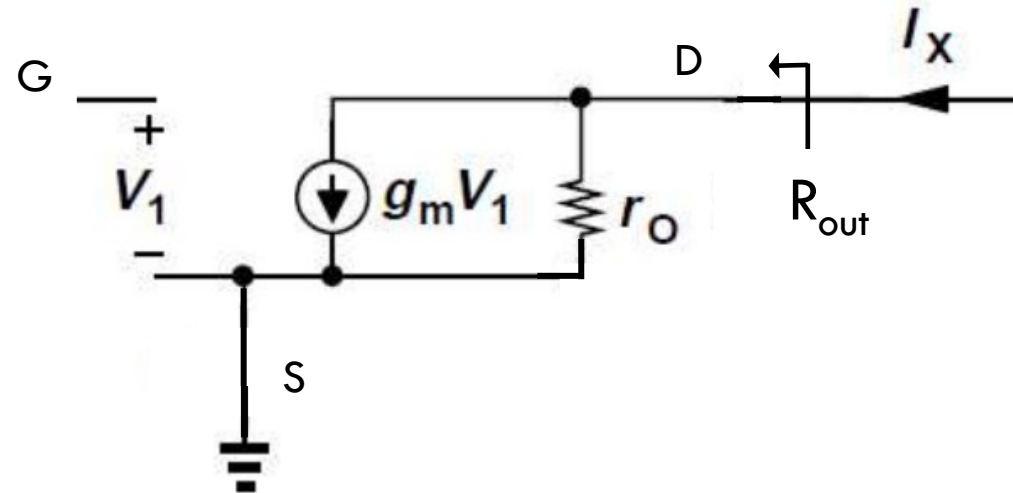
PARAMETERS OF MERIT FOR A CURRENT MIRROR



The output resistance of a voltage source shall be as small as possible. r_{out1} shall be as small as possible.

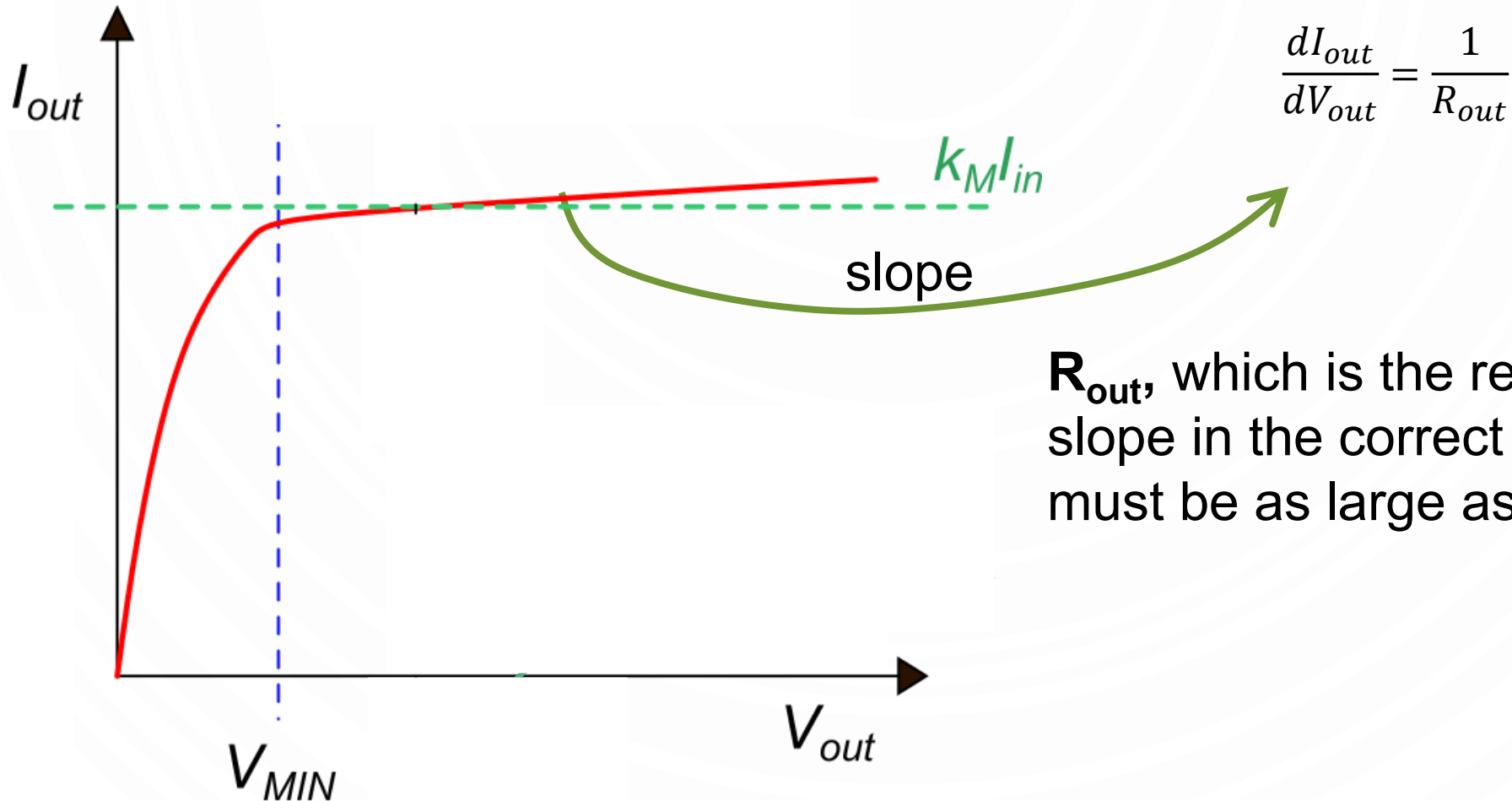
The output resistance of a current source/sink shall be as large as possible. r_{out2} shall be as large as possible.

Why?



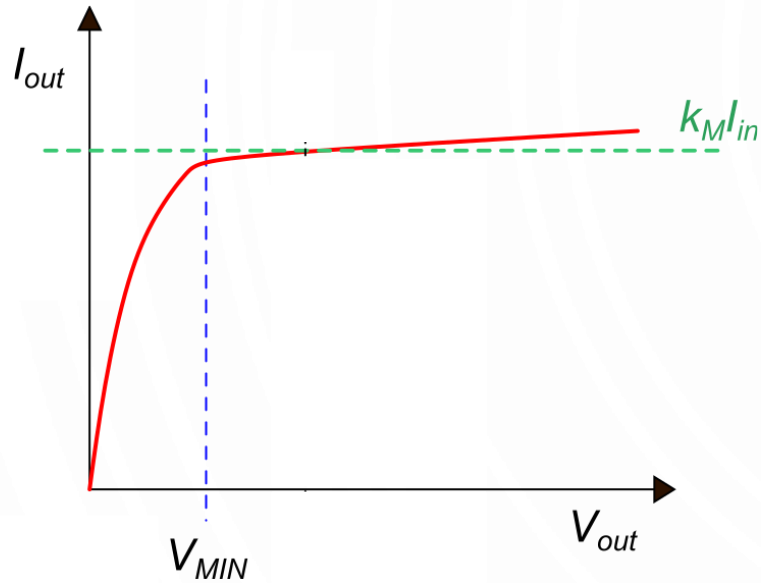
We neglect the capacitance between gate and source.

THE TYPICAL I_{OUT} VS V_{OUT} CHARACTERISTIC OF A CURRENT MIRROR



R_{out} , which is the reciprocal of the slope in the correct operation zone, must be as large as possible

THE CURRENT MIRROR VOLTAGE



current variation: $\Delta I_{out} = \Delta V_{out} \frac{1}{R_{out}}$

relative current variation:

$$\frac{\Delta I_{out}}{I_{out}} = \Delta V_{out} \frac{1}{I_{out} R_{out}}$$

Drain current with channel length modulation:

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} [(V_{GS} - V_T)^2] (1 + \lambda V_{DS})$$

The higher the output resistance, the more ideal the current mirror.

WHY DO WE NEED CURRENT MIRRORS?

Small reference currents can be generated in a central reference supply voltage/current generation block. These reference voltages and currents can be distributed to the whole system to generate larger currents by using the corresponding current mirror ratios k_M for each target circuit like amplifiers, op amps etc.

Current mirrors can also be used as active loads instead of resistors but this topic will be covered in the later weeks...

- Paralleling of BJTs
- Paralleling of MOSFETs

Please make a comparison of these two!

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.